02. Friday, June 21. **Topological spaces (Ch. 12)**

Definition 1: A **topology** on a set X is a collection of subsets $\mathcal{T} \subseteq \mathcal{P}(X)$ such that:

1) The sets $\emptyset$ and X are both in $\mathcal{T}$.

2) If $\{U_i \mid i \in \Lambda\} \subseteq \mathcal{T}$ then $\bigcup_{i \in \Lambda} U_i \in \mathcal{T}$. "$\mathcal{T}$ closed under arbitrary unions"

3) If $U_1, U_2 \in \mathcal{T}$ then $U_1 \cap U_2 \in \mathcal{T}$. "$\mathcal{T}$ is closed under finite intersections"

In this case, we say that the pair $(X, \mathcal{T})$ is a **topological space**.

The elements of $\mathcal{T}$ are called **open sets**.

Remark: For any collection $\mathcal{T} \subseteq \mathcal{P}(X)$, axiom 3 is equivalent to:

(3') For any $n \in \mathbb{N}$, if $U_1, \dots, U_n \in \mathcal{T}$ then $\bigcap_{i=1}^{n} U_i \in \mathcal{T}$.

Proof: (3') $\Rightarrow$ (3) is clear.

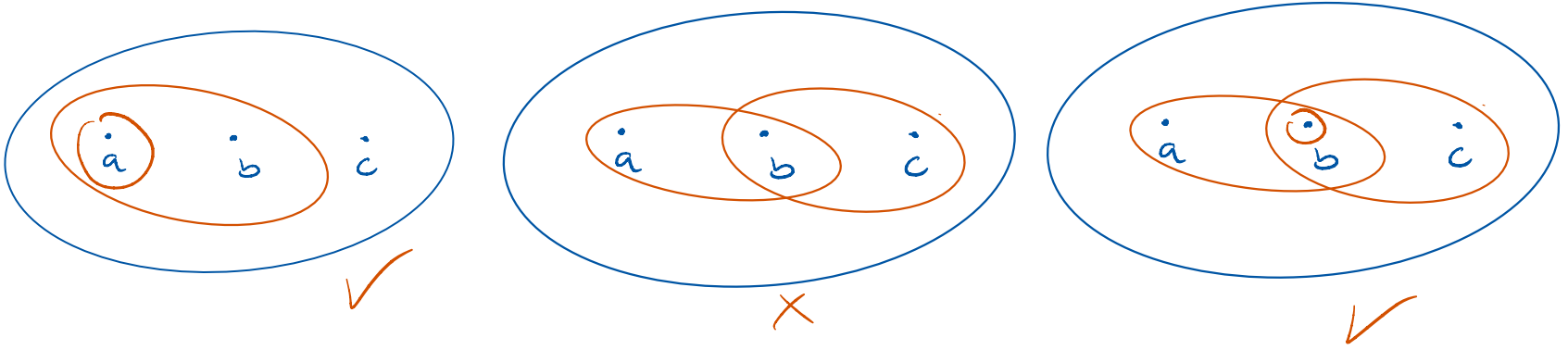
(3) $\Rightarrow$ (3') Exercise! (Just a refresh on induction) $\square$

Examples:

1) For any set X, the family $\mathcal{T}_{Triv} = \{\emptyset, X\}$ is a topology "The Trivial topology on X"

2) For any set X, the family $\mathcal{T}_{disc} = \mathcal{P}(X)$ is a topology "The discrete topology on X"

3) For the set $X = \{a, b, c\}$



✓ X ✓

4) For any set X, the family

$$\mathcal{T}_{fc} = \{U \subseteq X \mid X - U \text{ is finite}\} \cup \{\emptyset\}$$

is a topology, called the cofinite topology.

Motivating example: The standard topology of $\mathbb{R}$.

Notation: Let $x \in \mathbb{R}$ and $\varepsilon > 0$. Denote $B(x,\varepsilon) = \{y \in \mathbb{R} \mid |x-y| < \varepsilon\} = (x-\varepsilon, x+\varepsilon)$.

We define $T_{std} = \{U \subseteq \mathbb{R} \mid \forall x \in U, \exists \varepsilon > 0 \text{ such that } B(x,\varepsilon) \subseteq U\}$

This is a topology:

1) $\emptyset \in T$ is vacuously true. Also $\mathbb{R} \in T_{std}$ (given $x \in \mathbb{R}$, take any ε).

2) Arbitrary unions: Assume we have $U_i \in T_{std}\ \forall i \in \Lambda$. WTS $\bigcup_{i\in\Lambda} U_i \in T_{std}$.

Let $x \in \bigcup_{i\in\Lambda} U_i$. Then $x \in U_j$ for some $j \in \Lambda$. Since $U_j \in T_{std}$ and $x \in U_j$, we know $B(x,\varepsilon) \subseteq U_j$ for some $\varepsilon > 0$. So also $B(x,\varepsilon) \subseteq U_j \subseteq \bigcup_{i\in\Lambda} U_i$ □

3) Finite intersections: Let $U_1, U_2 \in T_{std}$. WTS $U_1 \cap U_2 \in T_{std}$.

Let $x \in U_1 \cap U_2$. Then $\exists\ \varepsilon_1, \varepsilon_2 > 0$ s.t. $B(x,\varepsilon_1) \subseteq U_1$, $B(x,\varepsilon_2) \subseteq U_2$. If we take $\varepsilon = \min(\varepsilon_1, \varepsilon_2)$, Then $B(x,\varepsilon) \subseteq B(x,\varepsilon_1) \cap B(x,\varepsilon_2) \subseteq U_1 \cap U_2$. □

Q: How useful is this notion at measuring nearness?

Recall: A sequence $x_1, \dots, x_n, \dots$ in $\mathbb{R}$ converges to $x \in \mathbb{R}$ if $\forall \varepsilon > 0 \ \exists N \in \mathbb{N}$ s.t. $|x - x_n| < \varepsilon \ \forall n \geq N$.

Exercise: Show that $x_n \longrightarrow x$ if and only if $\forall U \in \mathcal{T}_{std}$ with $x \in U$, $\exists N \in \mathbb{N}$ s.t. $x_n \in U \ \forall n \geq N$.

Exercise: Show that each $B(x, \varepsilon)$ is open.

Definition 2: Let X be a Top. space.

- A subset $C \subseteq X$ is **closed** if $X - C$ is open.
- A subset $N \subseteq X$ is a **neighborhood** of a point $x \in X$ if $\exists U \in \mathcal{T}$ s.t. $x \in U \subseteq N$.

Example: In $\mathbb{R}$,

- The interval (a, b) is open: Given $x \in (a, b)$, if we take $\varepsilon = \min(|x-a|, |x-b|)$, then $B(x, \varepsilon) \subseteq (a, b)$. Also $(-\infty, b)$ and (a, ∞) are open.
- The interval $[a, b]$ is closed. In fact $\mathbb{R} - [a, b] = (-\infty, a) \cup (b, \infty)$ is open (union of two open sets)
- The intervals $[a, b)$ and $(a, b]$ are neither open nor closed.